

Supplementary Information for De novo design of miniprotein antagonists of cytokine storm inducers

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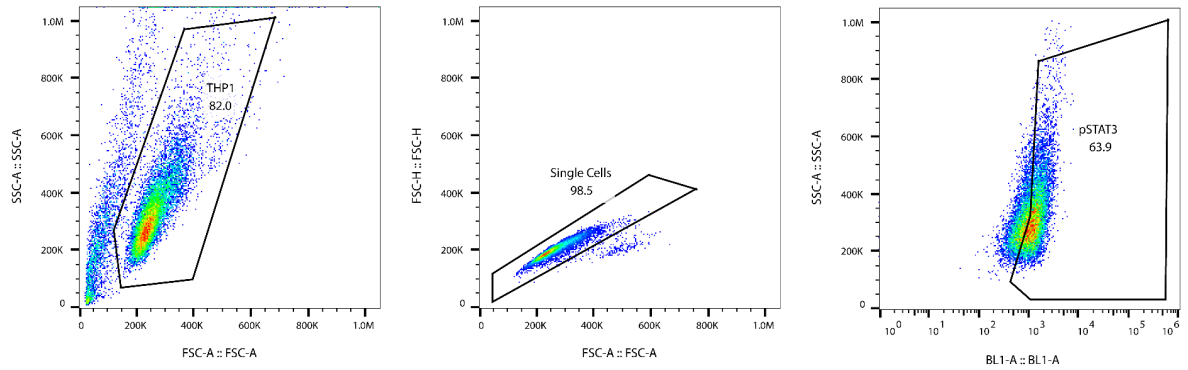
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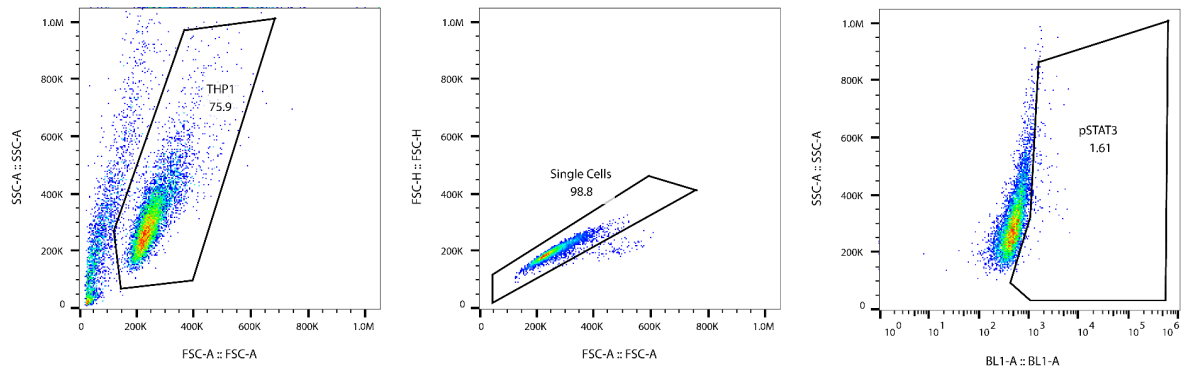
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Supplementary Figure 1. IL-6 signaling phospho flow plot. The represented samples were stimulated with 1.59nM of IL6 and pre-incubated without antagonist (Control), 100nM IL-6Rmb10 or 100nM GP130mb33.

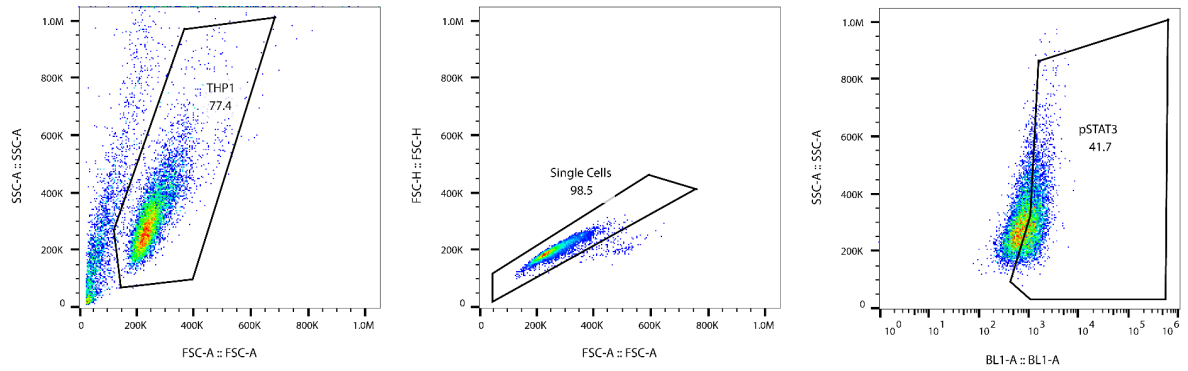
Control



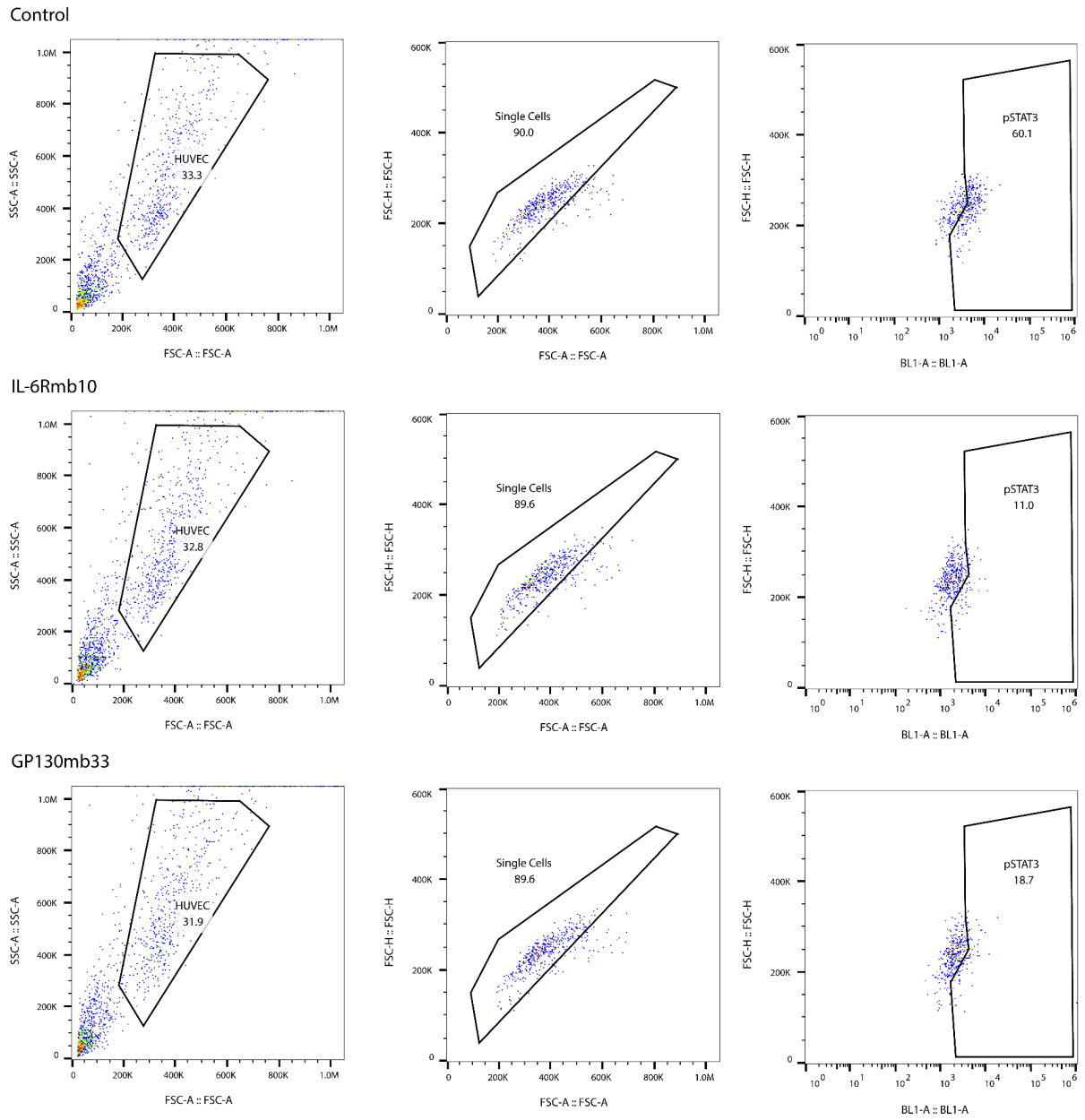
IL-6Rmb10



GP130mb33

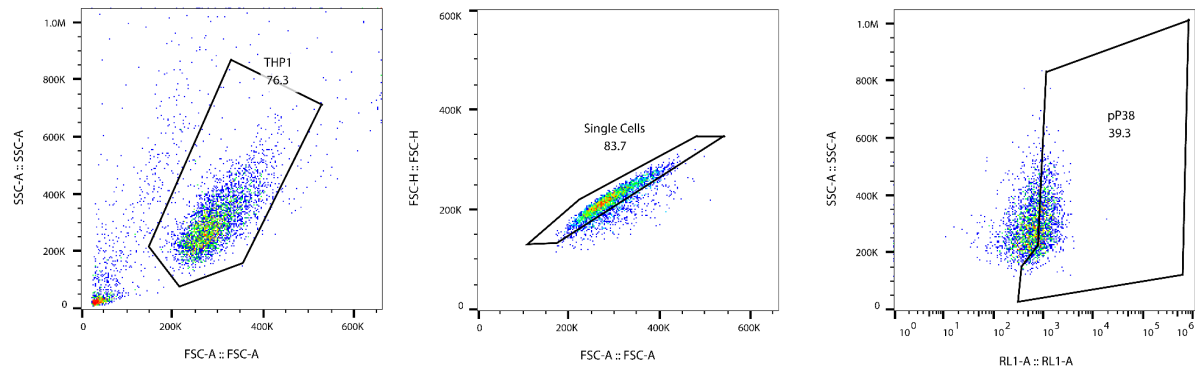


Supplementary Figure 2. IL-6 trans-signaling phospho flow plot. The represented samples were stimulated with 100nM of IL6 and pre-incubated with 200nM of soluble IL-6R and without antagonist (Control), 500nM IL-6Rmb10 or 500nM GP130mb33.

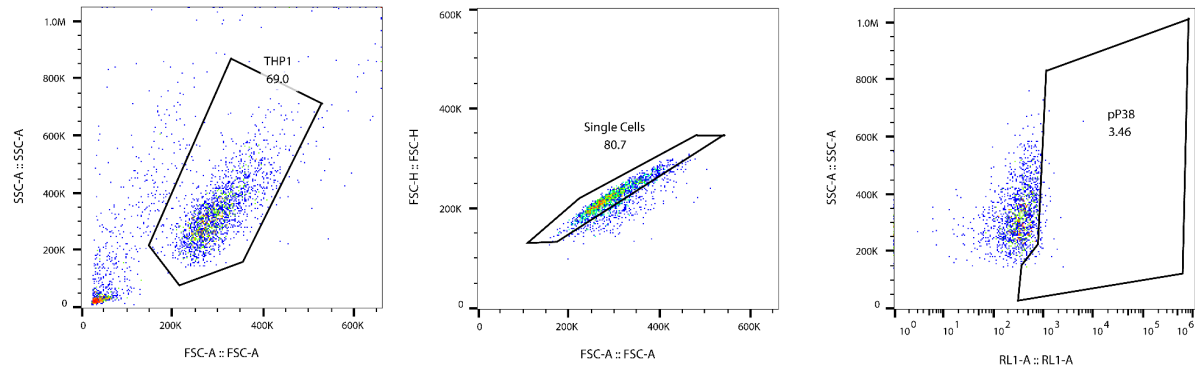


Supplementary Figure 3. IL-1 signaling phospho flow plot. IL-1 The represented samples were pre-incubated with 2uM of IL-1Rmb81 or PBS control, then washed and stimulated with 10nM IL-1.

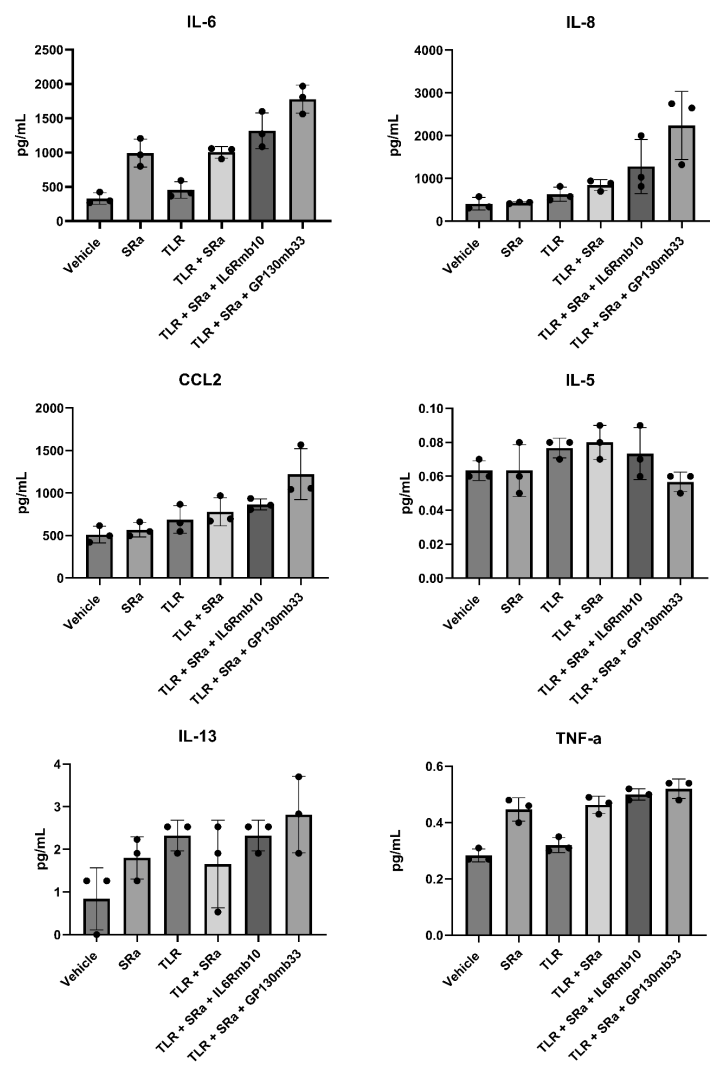
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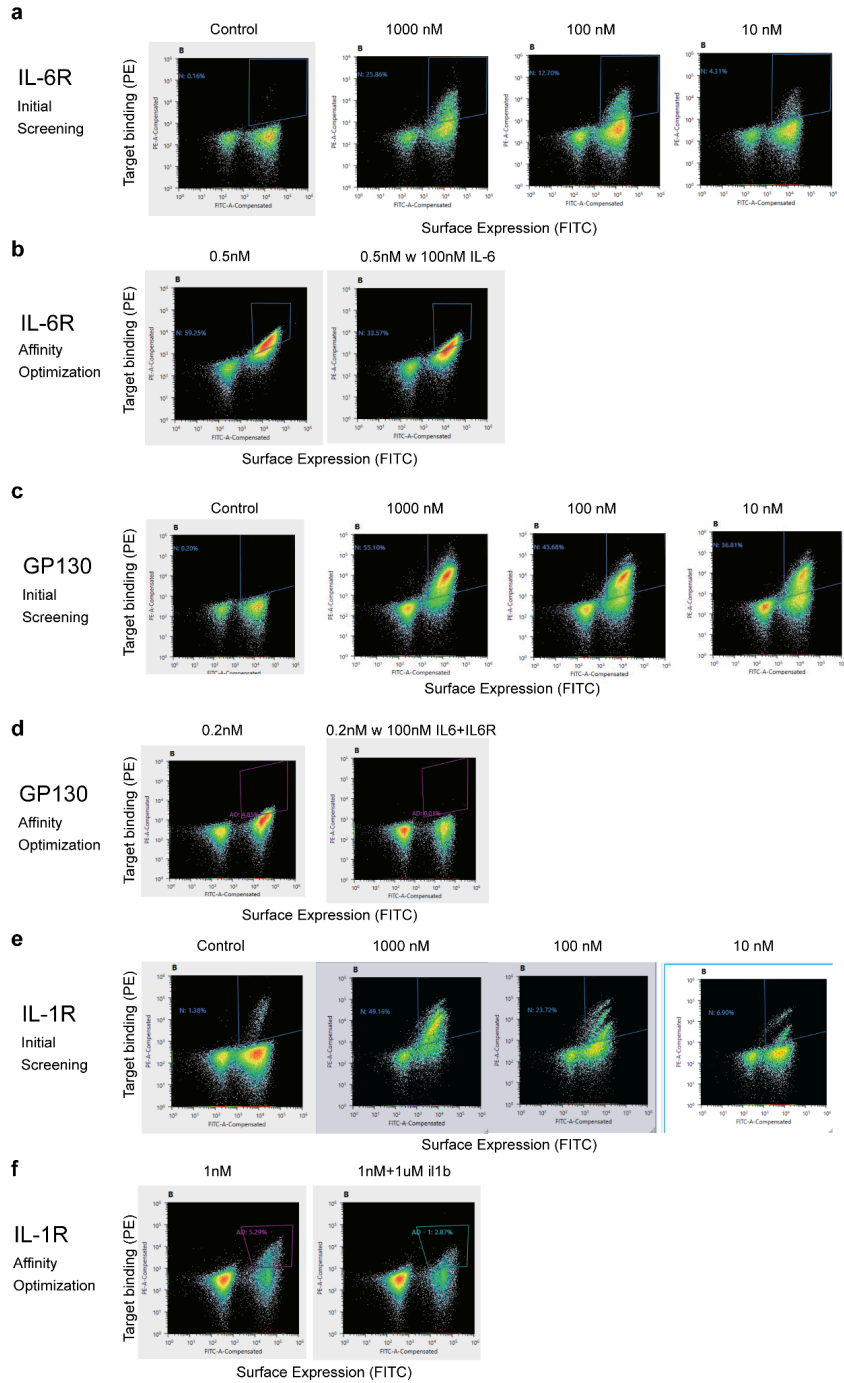
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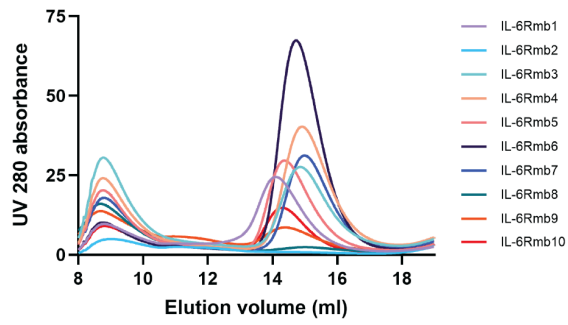
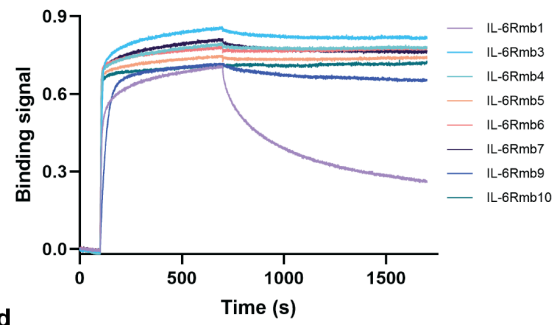
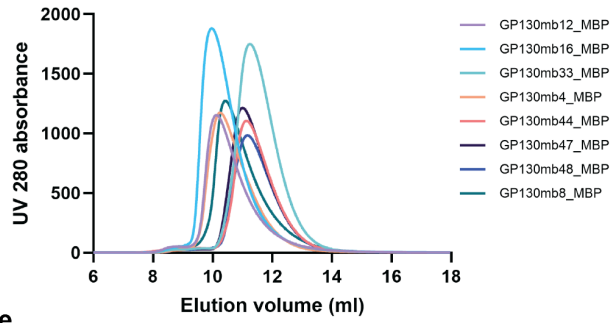
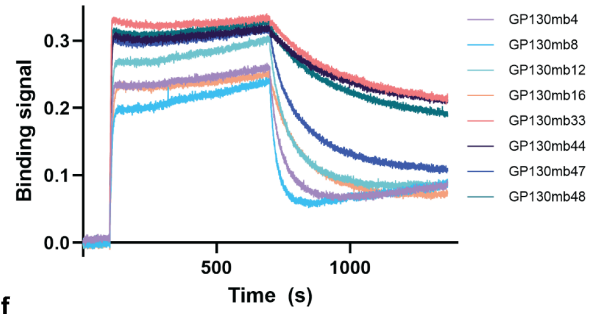
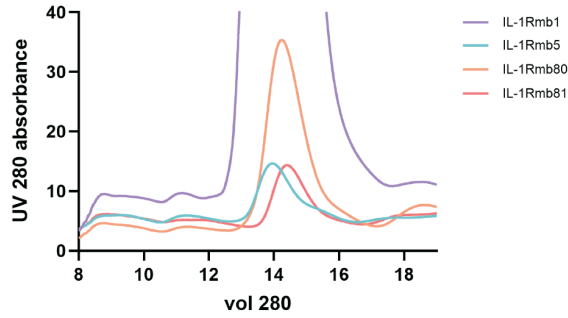
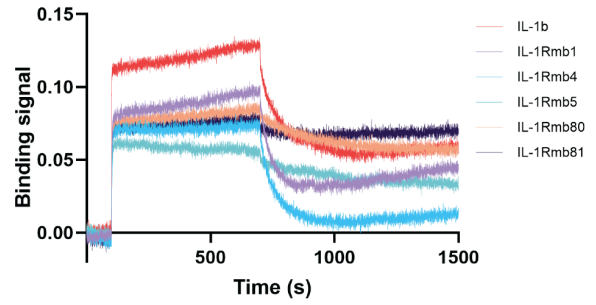
Supplementary Figure 4. Cytokine multiplex for organoids treated with TLR agonists or soluble IL-6R followed by IL-6Rmb6 and GP130mb33 inhibitors. Results were represented as fold change of cytokines in Day4 supernatant as compared to hCOs. Data are presented as mean values +/- SEM with biological replicates with a sample size of n=3.



Supplementary Figure 5. Yeast surface display screening for designed antagonists. **a**, Initial round affinity screening for IL-6R. **b**, Affinity optimization for top IL-6R hit (left) and competition of native IL-6 (right). **c**, Initial round affinity screening for GP130. **d**, Affinity optimization for top GP130 hit (left) and competition of native IL-6 and soluble IL-6R (right). **e**, Initial round affinity screening for IL-1R. **f**, Affinity optimization for top IL-1R hit (left) and competition of native IL-1b (right). Yeast cells displayed the designed proteins were incubated with biotinylated corresponding target protein for 30min followed by 2 round of wash with PBS with 1% BSA, then labeled with anti-cMYC-FITC antibody and SA-PE antibody (both 1:100 ratio). All control groups were yeast cells with the same labeling without adding target protein. The gating was based on the control groups.



Supplementary Figure 6. Characterization of designed minibinder candidates with SEC and BLI. **a**, SEC traces for combo-optimized IL-6R minibinders. **b**, BLI traces for combo-optimized IL-6R minibinders. 50nM biotinylated IL-6R were loaded in the SA sensor tip and incubated with 1000nM minibinder solution. **c**, SEC traces for combo-optimized GP130 minibinder candidates. **d**, BLI traces for combo-optimized GP130 minibinders. 50nM biotinylated GP130 were loaded in the SA sensor tip and incubated with 500nM minibinder solution. **e**, SEC traces for combo-optimized IL-1R minibinder candidates. **f**, BLI traces for combo-optimized IL-1R minibinders. 50nM biotinylated IL-6R were loaded in the SA sensor tip and incubated with 500nM minibinder solution. For **a-f**, all protein constructs in the plot were Ni-NTA column purified from E coli culture elutions. For **c-d**, the GP130 minibinders were genetically fused with an MBP tag to amplify the BLI binding signal. For **a,c,e**, the SEC traces were collected from in S75 Increase column (Cytiva).

a**b****c****d****e****f**

Supplementary Figure 7. Cytokine signaling inhibition effects of designed antagonists. a,

Phosphorylation flow assay on STAT3 signaling. THP-1 cells were titrated with increased

concentrations of IL-6 alone or with 100 nM IL-6Rmb6 or IL-6Rmb10. **b,** Phosphorylation flow

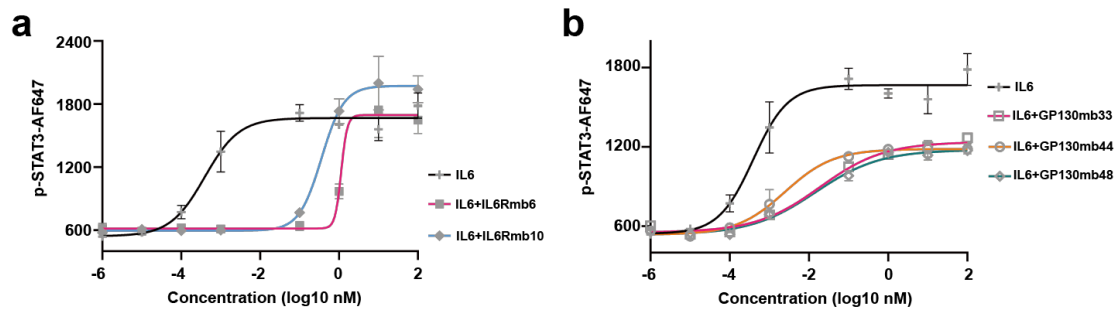
assay on STAT3 signaling. THP-1 cells were titrated with increased concentrations of IL-6 alone

or with 100 nM GP130 antagonists. For **a-b**, the mean values were calculated based on the

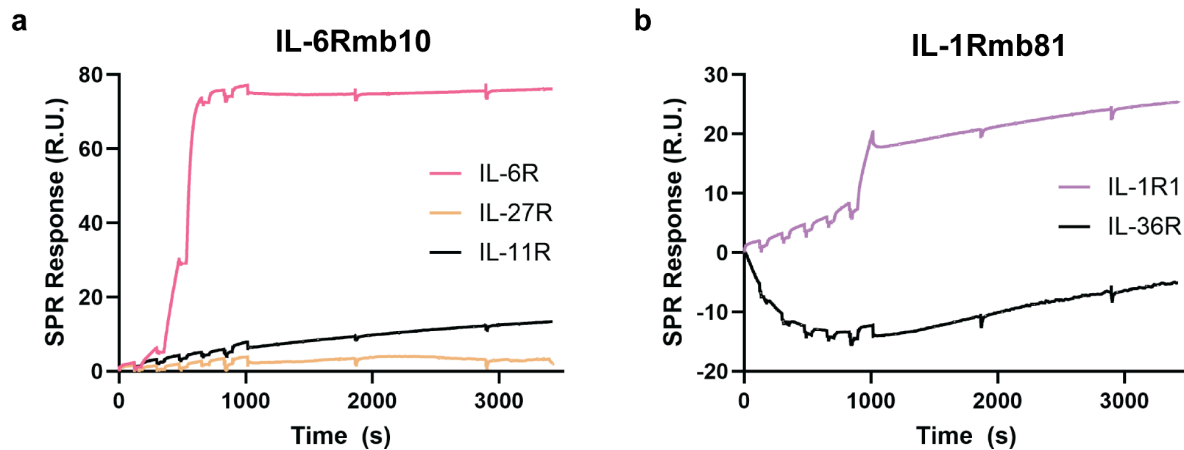
triplicates and error bars represent standard deviations. The IC₅₀s were measured by non-lin fit

using GraphPad Prism 8.0.2. Data are presented as mean values +/- SEM with biological

replicates with a sample size of n=3.



Supplementary Figure 8. Crossreactivity of designed antagonists to homologous receptors. **a**, SPR binding curve of IL-6Rmb10 to IL-6R and homologous receptors IL-27R and IL-11R. Fc-tagged receptor protein at 5 ug/ml concentration was loaded into Biacore ProteinA-chip, and incubated with increased concentration IL-6Rmb10 binder solution (6 steps of 8x dilutions from 1000 nM) with single-cell kinetics curve measurement. **b**, SPR binding curve of IL-1Rmb81 to IL-1R1 and homologous receptors IL-36R. Fc-tagged receptor protein at 5 ug/ml concentration was loaded into Biacore ProteinA-chip, and incubated with increased concentration IL-1Rmb81 binder solution (6 steps of 8x dilutions from 1000 nM) with single-cell kinetics curve measurement. The data was obtained in Biacore™ 8K SPR system.



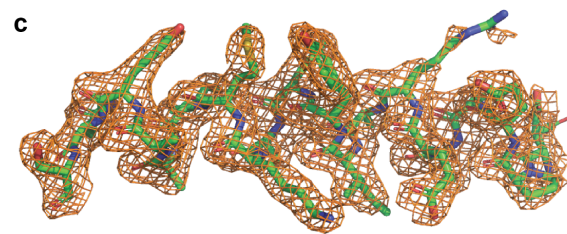
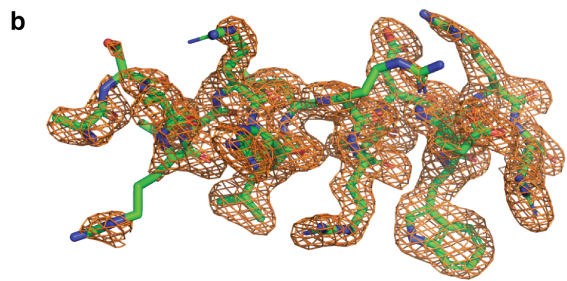
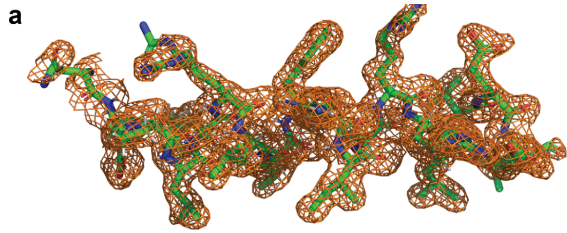
Supplementary Figure 9. Complex structure of GP130mb44 in complex with GP130.

Alignment of designed model of GP130mb_original (before optimization) in complex with GP130 to experimentally determined crystal structure for GP130/GP130mb44 complex. The rmsd is measured by aligned complex rmsd using pymol2.

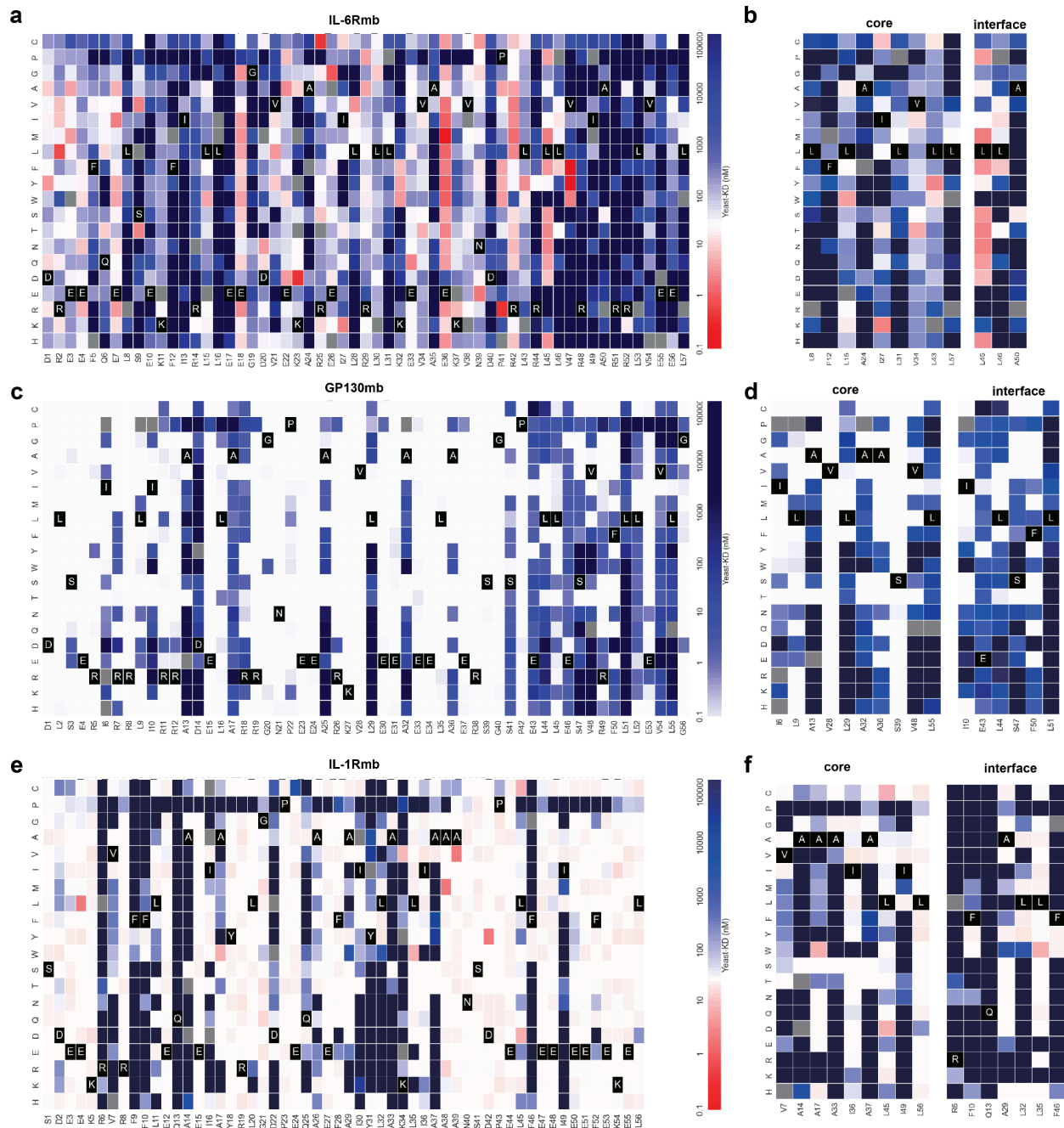
GP130mb44 (Crystal complex)
GP130mb_original (Design model)
Complex RMSD=0.9A



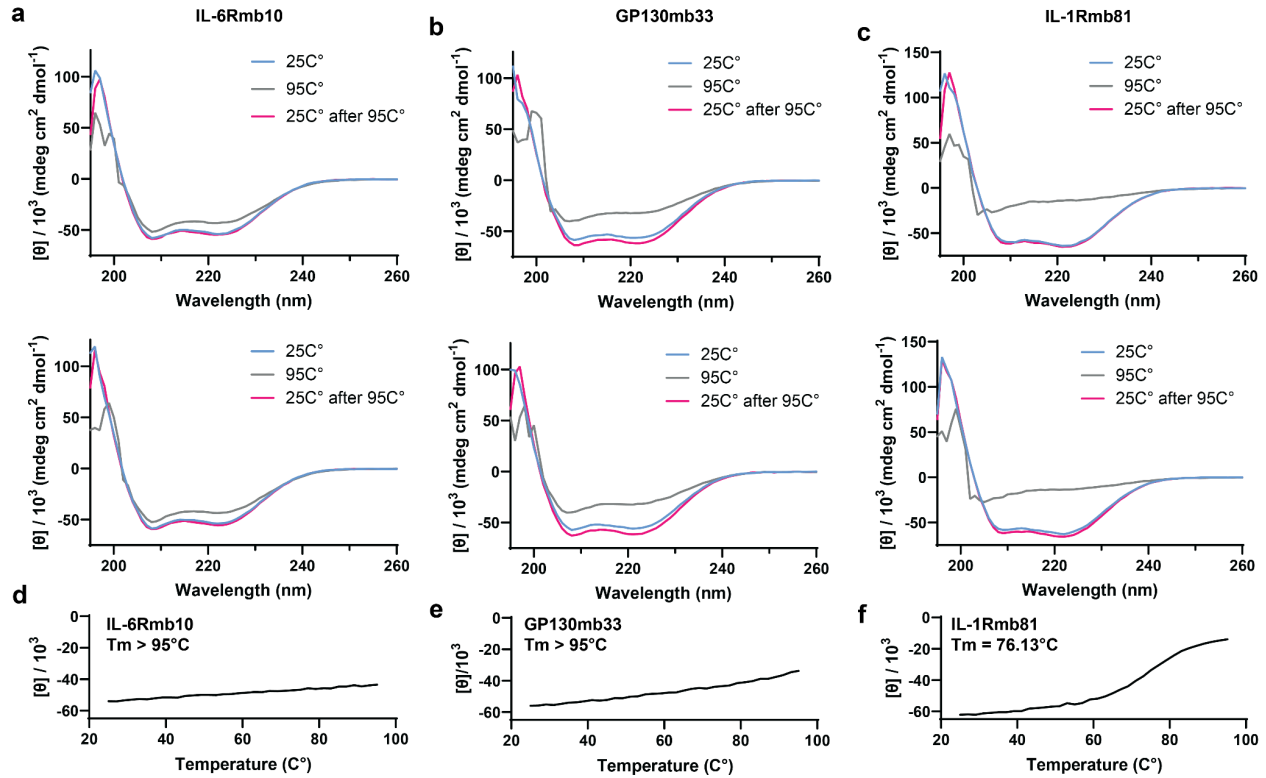
Supplementary Figure 10. Electron density map for crystal structures. 2mFo-DFc electron density map contoured at 1 sigma. **a**, IL6Rmb, chain A, resi 47-65. **b**, gp130mb complex, chain A, resi 2-20. **c**, IL1Rmb, chain A, resi 22-42.



Supplementary Figure 11. SSM heatmap for originally designed minibinders. **a**, Full mutation heatmap for IL-6Rmb. **b**, Zoomed mutation profile in core and interface of the designed minibinder. **c**, Full mutation heatmap for GP130mb. **d**, Zoomed mutation profile in core and interface of the designed minibinder. **e**, Full mutation heatmap for IL-1Rmb. **f**, Zoomed mutation profile in core and interface of the designed minibinder. The Sorting Concentration where 50% of the expressing cells were collected (SC50) for each of the variants was calculated from the yeast-display cell-counts before and after the titration. For designs that did not appear in the expression sort or that could not be reliably assigned a SC50 value, a grey cell is shown.



Supplementary Figure 12. CD spectrum for designed minibinders. a-c, Replicated circular dichroism spectrum measured at different temperatures. **d-f,** T_m curves measured by recording the circular dichroism signals at increased temperature from 25°C to 95°C. T_m values were measured by calculating the inflection point of the sigmoidal fitted curve using GraphPad Prism. Source data is provided as a Source Data file.



Supplementary Tables

Supplementary Table 1. Statistics for the crystal structure determination

	IL-1Rmb80	IL6Rmb6	GP130mb33	GP130mb44
Resolution range (Å)	36.9 - 1.96 (2.08 - 1.96)	33.24 - 1.48 (1.533 - 1.48)	29.96 - 3.303 (3.421 - 3.303)	48.37 - 1.87 (1.937 - 1.87)
Space group	C 1 2 1	P 41 21 2	I 21 3	P 32 2 1
Unit cell	83.3 55.2 53.9 90 103.54 90	47.0 47.0 74.2 90 90 90	189.5 189.5 189.5 90 90 90	90.8 90.8 61.4 90 90 120
Total reflections	117319 (19264)	28917 (2767)	32821 (3176)	46621 (4714)
Unique reflections	16949 (2831)	14455 (1384)	17077 (1660)	24415 (2428)
Multiplicity	6.9 (6.8)	2.0 (2.0)	1.9 (1.9)	1.9 (1.9)
Completeness (%)	98.15 (98.29)	99.89 (99.71)	99.45 (96.60)	99.85 (99.92)
Mean I/sigma (I)	9.75 (0.84)	13.97 (0.83)	11.16 (0.92)	6.74 (0.94)
Wilson B-factor (Å ²)	38.19	24.15	109.37	31.83
R-merge	0.11 (2.22)	0.01878 (0.6176)	0.076 (0.85)	0.15 (0.70)
R-meas	0.12 (2.4)	0.026 (0.87)	0.108 (1.20)	0.21 (0.99)
R-pim	0.04372 (0.9199)	0.01878 (0.6176)	0.07636 (0.848)	0.1464 (0.697)
CC1/2	0.999 (0.46)	0.999 (0.539)	0.994 (0.289)	0.779 (0.369)
CC*	1 (0.794)	1 (0.837)	0.999 (0.669)	0.936 (0.734)
Reflections used in refinement	16897 (2812)	14455 (1384)	17059 (1646)	24413 (2428)
Reflections used for R-free	844 (140)	1445 (139)	1722 (164)	1986 (202)

R-work	0.1995 (0.3480)	0.2041 (0.3443)	0.2003 (0.3705)	0.1996 (0.3027)
R-free	0.2368 (0.4131)	0.2289 (0.3552)	0.2393 (0.4595)	0.2331 (0.3699)
Number of non-hydrogen atoms	1977	641	4133	2310
macromolecules	1941	574	4070	2081
ligands	6	0	61	14
solvent	30	67	2	215
Protein residues	232	64	509	256
RMS(bonds) (Å)	0.002	0.006	0.003	0.012
RMS(angles) (°)	0.46	0.92	0.57	1.58
Ramachandran favored (%)	100.00	98.39	97.38	98.73
Ramachandran allowed (%)	0.00	1.61	2.62	1.27
Ramachandran outliers (%)	0.00	0.00	0.00	0.00
Rotamer outliers (%)	1.00	0.00	4.65	1.77
Clashscore	1.33	2.79	4.79	1.69
Average B-factor (Å ²)	44.80	29.77	120.54	36.96
macromolecules	44.69	28.50	119.84	36.23
ligands	66.12	-	168.35	55.87
solvent	47.70	40.63	76.91	42.81

Supplementary Table 2. Sequences and mutations of the designed minibinders

Target	Design_name	Final Sequence	Number of Mutations
IL-6R	IL-6Rmb_original	DREEFQELSEKFIRLLEEGLDVEKAREILRLLKEVAE KVNDPRLRLLVRIARRLVEEL	-
IL-6R	IL-6Rmb6	DREEFQWLVEEFIRVLEGLDVEKAREILRLLKEVAE KVNDPLLRLLFRIARRLVEEL	7
IL-6R	IL-6Rmb10	DREEFQELVEEFIRVLEGLDVEKAREILRELKEVAE KVNDPLLRLQLFRIARRLVEEL	8
GP130	GP130mb_original	DLSEIRRLIRRADELARRGNPEEARKVLEEAEELA ERSGSPELLESVRFLLEVLG	-
GP130	GP130mb33	DISERFRRLMRRADELARRGNPEEARKVLEEAEELM ERYGSPELLESVRMLLEVLG	6
GP130	GP130mb44	DISERMRLMRRADEWARRGNPEEARKVLEEAEELA EQYGSPELLESVRMLLEVLG	7
IL-1R	IL-1Rmb_original	SDEEKVRFFLEQAEIAYRLGDPEQAEFAIYLAKLI AAANSDPPELFEEIEEFEKEL	-
IL-1R	IL-1Rmb80	SDEEEKVRFYLEQAEIHYRLGDPEAAERAIYLAKMI AAENSDPPELFEEIEEFEKEL	8
IL-1R	IL-1Rmb81	SDEEEKVRFYLEQAEIHYRLGDPEAAEMAIYLAKMI AAENSDPPELFEEIEEFEKEL	8

Supplementary Table 3. Fitting table for IL-6 STAT3 signaling inhibition assay

Parameters	Control	GP130mb33	IL-6Rmb10
Top	77412	48432	68531
Bottom	1050	719.8	730.1
LogIC50	-0.536	-0.074	1.337
HillSlope	1.603	1.323	1.628
E _{max}	76362	47712.2	67800.9
E _{max} _foldchange	1.000	0.625	0.888
EC50	0.291	0.843	21.740
EC50_foldchange	1.000	2.894	74.657
Span	76362	47713	67801
Degrees of Freedom	44	44	44
R squared	0.971	0.993	0.952
Sum of Squares	1.51E+09	1.26E+08	7.06E+08
Sy.x	5866	1692	4007

Supplementary Methods

Supplementary Method 1. Method for signaling assay described in Supplementary Fig. 7:

For STAT signaling, THP-1 human monocyte cell line cells were grown in RPMI complete media (Corning), then 100k cells/well were incubated with 100 nM antagonist (GP130mb) for 10 min @ 37 C (no wash), then stimulated with hIL-6 (0.01pM-100nM, PeproTech) for 45min @37 C prior to fixation with 1.6% paraformaldehyde (ThermoFisher) for 10 minutes at room temperature. Cells were pelleted, re-suspended in methanol and stored at -20 C until analysis. Cells were washed in FACS buffer (PBS pH 7.4 with 2% FBS and 2mM EDTA) and stained with antibodies against 1:50 (v/v) STAT3 pY705 AF647 (BD) for 1 h at room temperature. Fluorescence intensity of pSTAT staining was analyzed using a CytoFlex flow cytometer (Beckman Coulter).